2024 IEEE Pune Section International Conference (PuneCon)
Pune, India. Dec 13-15, 2024

Revolutionizing the Future of Automated Subjective Answer Sheet Evaluation System with Machine Learning and LLMs

[1]Balla Aarathisree
Department of CSE (Data Science)
Pragati Engineering College
Andhra Pradesh, India
aarathisree.1535@gmail.com

[2]Sujit Sarkar
Department of Computer Engineering (Data Science)
Army Institute of Technology
Pune, India
sujit.sarkar6112@gmail.com

[3]Bhanu Durgesh Dammala
Department of CSE (Data Science)
Pragati Engineering College
Andhra Pradesh, India
dammalabhanudurgesh@gmail.com

[4]Mrityunjoy Panday
Society for Data Science
mrityunjoy@gmail.com

[5]Neha Sharma
Society for Data Science
nvsharma1975@gmail.com

2024 IEEE Pune Section International Conference (PuneCon) | 979-8-3315-2782-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/PuneCon63413.2024.10895748

***Abstract* -- Human evaluation of handwritten answer sheets is very inefficient, inaccurate, and potentially biased. This research work serves the purpose of addressing these drawbacks by developing an automated grading system that leverages advanced OCR technologies and machine learning algorithms. The system, utilizes Google Cloud Platform (Cloud Vision API) and Gemini 1.5 Pro for Optical Character Recognition, to process large capacity of answer sheets with variation in handwriting styles more accurately than the traditional methods. This leads to a significant increase in text extraction accuracy and grading reliability. Therefore, as comparison with the existing solutions demonstrates, the new system has an increased level of accuracy and efficiency, reducing human error and fairly assessing student performances. Thus, implementation of changes in traditional grading system may influence the feasibility of this work, giving educators the opportunity to engage more time in instructional activities.**

Index Terms -- Large Language Models, LLM, Optical Character Recognition, OCR, Automated Grading Systems, Natural Language Processing, NLP, Prompt Engineering

I. INTRODUCTION

Evaluating handwritten answer sheets continues to be a constant challenge in modern education, troubled by issues of inaccuracy and inefficiency. In addition, there is a chance of potential biases during manual correction that increase its deficiency. Traditional paper-n-pen tests need coordination and physical labor, which is tedious, and prone to human mistakes in various stages [1][2]. The educational systems consist of large volumes of student work that not only require a vast period for evaluation but also to overcome these challenges. Nevertheless, the experience of human scorers is being restricted to evaluation thereby reducing the time available for other instructional activities and student engagement [3]. This research work proposes the development of an Automated Grading System to acknowledge these challenges. The system includes an answer sheet evaluation and ranking system leveraging advanced Optical Character Recognition (OCR) technologies and machine learning algorithms. Implementing these technologies enhance efficiency, accuracy, and fairness of traditional evaluation of answer sheets.

The labor-intensive method of evaluating student answer sheets is prone to manual error. These manual errors, inconsistencies and potential biases are observed due to the extensive navigation of evaluators through a variety of handwriting styles and formats [4]. Human evaluators experience tiredness due to the evaluation of large volumes of answer sheets, thereby increasing the chances of errors.

Educational institutions witness a serious administrative burden due to these challenges. These drawbacks highlight inefficiency delays in the feedback cycle for students. The Google Cloud Vision API from Google Cloud Platform and Gemini 1.5 Pro for Optical Character Recognition are combined to reduce human intervention in the text extraction process from handwritten answer sheets. A combination of OCR technologies and machine learning algorithms is implemented assess the extracted text from student answer sheets with utmost accuracy. In addition to it, Natural Language Processing (NLP) techniques are incorporated to assess the semantic content of student responses. Therefore, a robust mechanism of evaluating student answer sheets is acknowledged by implementing all these methodologies to capture the complicated aspects of human judgment.

The use of Google Cloud Vision API provides a highly reliable OCR capability, proficient at handling a variety of handwriting styles and ensuring high accuracy. Gemini 1.5 Pro further enhances the capability of text extraction by providing additional tools for text analysis and error correction. A highly accurate and rapid evaluation of huge amount of student answer sheets is achieved by combining all these technologies. Advantages of this system over the existing manual and semi-automated grading methods. The Automated Answer Sheet Evaluation and Ranking System gives fewer overheads than any method [5][6]. In the first place, it expedited student feedback by orders of magnitude in decreasing time to grade large piles of answer sheets. Moreover, the addition of a superior performing OCR and machine learning technologies nullifies the remnants of manual grading problems which also lead to better text extraction and enhancing accurate reading. The system lastly verifies the ability to evaluate all students with equitable and fairness score, as it capable to give a consistent review from student perspectives without bias can increase result of grading more fairly.

The proposed solution erases the current drawbacks involving the repetitiveness and time consumption during the process of manual assessment. Besides, it creates an efficient and time-saving evaluation process and makes it normal to contribute more time to what counts most: students and instruction. It enhances the general quality of education while progressing the enhancement of the provision of educational resources. Thus, the conventional grading practices are transformed to modern evaluation practices through the implementation of the Automated Answer Sheet Evaluation and Ranking System. The proposed system utilizes complex OCR and machine learning techniques to automate the process of manual grading of multiple student assessments with ease and accuracy. This research paper bears vast implications to the specific possibility of AI and Machine learning in Educational Assessments paving new paths towards next generation educational practices [9][10].

This advancement promises to elevate the quality of education, providing a more reliable and equitable system for evaluating student performance.

II. LITERATURE REVIEW

Traditional methods result into laborious grading, unfair evaluation comparison between students, and are prone to human errors. Whereas, Automated Grading System help in great reduction of the manual interaction but with a faster and highly accurate checks. Ms. Macha Babitha el. Al. states that Numerous automated grading programs are capable of analysing test papers and comparing and examining candidates' writing. Artificial intelligence (AI) technologies can easily recognise letters, numbers, and special symbols. These AI tools also have the advantage of learning like humans [1]. Thanks to machine learning, software will not repeat an issue if a system fault occurs and is assessed and fixed by a teacher. Some of these technologies can evaluate answer sheets in as low as 90% less time than teachers, demonstrating their speed. Consequently, findings will be made available shortly after the exam is completed, helping the institutions overcome their evaluator deficit and saving time and effort [2].

Madhavi Kulkarni et al. proposed a system that could modernize the process of evaluation of handwritten answer sheets by effectively incorporating technologies like Natural Language Processing and Optical Character Recognition with cosine similarity analysis. Their approach increases the accuracy and efficiency of the assessment process through improvements in the transcription, understanding, and comparison of handwritten responses. The system starts by the input of handwritten answer sheets which may be in the form of scanned documents or images taken using cameras or mobile devices [3]. These images are processed to extract the handwritten content and convert it into machine-readable text, using the OCR technology. In the process, the OCR recognizes text, segments characters, and extracts features. This is followed by tokenizing the extracted text, using much more advanced techniques like the BERT tokenizer, to be converted into word embeddings to further process it.

The hybrid system approach is implemented in the evaluation of the written answers by incorporating BERT-based similarity measurement and dissimilarity analysis by means of WordMoverDistance. [4] The student and teacher responses are encoded in vector forms through the Word2Vec model. The BERT-similarity is then mapped to the marks similarity conversion structure to allocate partial marks. WMD calculates the distance between responses, contrasted with a mark–distance conversion system that is keyed for 30% of all grades. This has an added advantage of maximizing the assessment system whereby students can access final scores on the same platform.

A.K. . Maya et. al. suggest that the Automated Grading System gives an ease for being useful in educational system where all are evaluated on same grounds as traditional methods [5]. Optical Character Recognition (OCR) and machine learning in grading is adopted for making the educational assessment more accurate, efficient, and fair. This literature suggests the searched alternatives applied in different methods of automatic answer sheet grading and ranking [5].

Hussein et. al. highlight that the need for automated systems in education is increasingly evident as traditional methods of grading are time-consuming, inequality in evaluation, and prone to human error [6]. The authors explore the existing research, methodologies, and technologies employed in the development of automated answer sheet evaluation and ranking systems.

Due to the ambiguity of natural language, subjective questions evaluate students' knowledge and skills in an open-ended way,

making automated evaluation difficult as mentioned in the work done by M. F. Bashir et. al. [7]. In order to investigate a machine learning and natural language processing strategy, this study uses methods such as tokenization, lemmatization, TF-IDF, and word2vec. Evaluation metrics that are utilized include accuracy and F1-score. A two-phase approach is suggested: first, using similarity metrics, compare student responses to a model response; second, train a model to assess responses on its own. The research tackles issues such as different word lengths, synonyms, and the requirement for enhanced datasets and hyper-parameter adjustments.

As per A. D. Alrehily et. al. the assessment system architecture consists of many modules that analyze student answers against reference answers. There are four modules: pre-processing, keyword expansion, matching, and grading [8]. The pre-processing module standardizes the input by converting both student and reference responses to lowercase and deleting stop words, punctuation, and prepositions. The module uses natural language processing techniques from the NLTK toolbox, such as tokenization and part-of-speech tagging, to clean up the replies before further analysis. To assess accuracy, the technology compares student replies directly to reference answers from a pre-approved shop. This modular method is intended to improve the accuracy and efficiency of grading subjective and objective exams.

III. METHOD AND IMPLEMENTATION

The methodology for the "Automated Answer Sheet Evaluation and Ranking System" involves several crucial stages for accurate text extraction and analysis. Bashir.et.al. in his

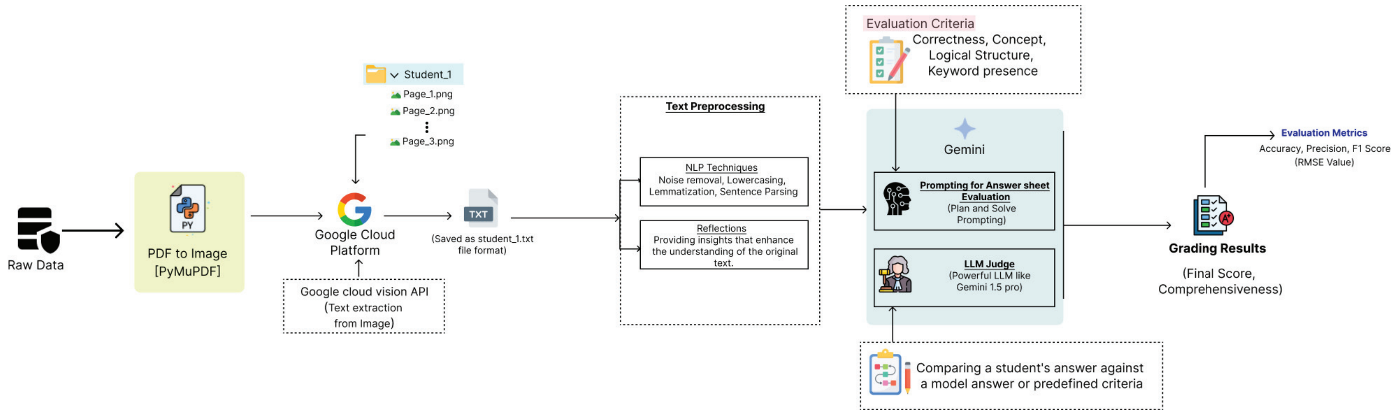

Fig. 1. Model Architecture

paper proposed that Pytesseract OCR was employed to extract text from answer sheets [7]. However, this approach proved impractical and non-scalable due to its limitations with large datasets and complex handwriting styles. Consequently, Google Cloud Vision API is adopted, offering greater accuracy and scalability for OCR tasks.

The multi-faceted evaluation criteria show some depth in approach, incorporating various factors such as keyword presence, conceptual accuracy, and textual consistency. However, the system lacks rigorous validation methods, particularly in ensuring consistency with human graders. To address this limitation, it is essential to conduct comprehensive validation, which studies comparing automated scores with those given by experienced educators. This would involve a detailed analysis of inter-rater reliability and the identification of any discrepancies between automated and human grading. Implementing such validation measures would enhance the credibility and acceptance of the automated system in educational settings, ensuring it meets the standards of fairness and accuracy expected in academic assessments.

The complete architecture of Automated Grading System is depicted in Figure 1. The process starts by converting scanned PDFs into images by running a Python script, which iterating through each folder and look into every image file (. png, jpg, jpeg).

The process starts by deploying OCR to read all pages of the scanned documents, making sure that it reads multiple text orientations and works at high accuracy mode. The only thing that needs to be done before comparing the texts, is to clean all extracted text removing irrelevant characters, in order to prepare answer sheets for accurate comparison. After cleaning, the extracted answers are judged using predefined criteria of evaluation provided by the subject expert. The assessment includes mapping of answer scheme to the answers given by the students. The entire paper is scored out of the total marks for that subject, based on the accuracy and quality of answers.

To create a robust and scalable solution, a Google Cloud Platform Project is setup. From the platform, Google Cloud Vision API is used to extract the text from the images. Simultaneously, Gemini 1.5 pro is setup to authenticate the API key for the Large Language Model (LLM) and the system environment variables. Besides, in Google Cloud Console, a service account is setup to permit the secured access to the application with Gemini 1.5 pro API.

The text is extracted using Google Cloud Vision API and saved in a .txt file to create a data structure that can be used for further analysis. Then NLP techniques are used to preprocess the text to make it ready for the analysis. The identified inconsistencies, mistakes, and uncertainties were handled using the NLP techniques, which helped in enhancing the correction process by Noise removal, Lemmatization, Lowercasing, Sentence parsing.

The quality of text created after preprocessing using NLP techniques was not that accurate and therefore Reflections were tried to further preprocess the text to make it more precise. Reflections are used for critical evaluation of the NLP techniques being used, such as how well it performs in extracting meaningful information from text, and identifying areas where improvement can be made.

Once the extracted text is processed and ready, multiple approaches were tried for evaluating the answer sheets and grading as per the scheme. In this paper, "Plan-and-Solve prompting" and "LLM as a Judge" methods have been adopted and are explained below:

A. *Plan and Solve Prompting*

For multi-step reasoning, few-shot chain-of-thought (CoT) prompting is used, which manually constructs step-by-step reasoning examples, allowing LLMs to develop clear logical processes and enhance the accuracy of the task, as suggested by L.Wang et.al [10]. To save human effort, Zero-shot-CoT joins the target issue statement with "Let's think step by step" as an input prompt for LLMs. Though Zero-shot-CoT has its success, it has some calculation problems, missing-step errors, and semantic misunderstandings. To address these issues, Plan-and-Solve (PS) Prompting method is suggested, which breaks the work into lower subtasks before working on them. Following are the steps used in Plan-and-Solve (PS) Prompting method:

Step 1: Prompting for Answer Extraction

To ensure accurate text extraction from documents, OCR technology is utilized to process all pages, enabling the detection and handling of text in various orientations. The high accuracy mode of the OCR engine is utilized to ensure precision in the extracted text. Additionally, text cleaning techniques are implemented to remove extraneous characters, thereby enhancing the clarity and usability of the extracted content.

Step 2: Facilitate Independent Evaluation

This scoring method provides a balanced evaluation of student understanding. Instead of restricting to the presence of keywords it acknowledges their ability to explain concepts accurately and structure their answer logically. It promotes a deeper understanding of the topic rather than just memorizing key terms.

B. Large Language Models as a Judge

When using an LLM as a judge identification of the evaluation criteria is required. This can be determination of what is required for the evaluation, assessment of hallucination, or any other characteristics.

A set of crafted prompts are required, which need to be guided for the evaluation as shown in figure 2. The prompt template should clearly define the which are the factors that are required from the first prompt and the LLMs answer to accurately evaluate the responses. For this Research project Gemini LLM model is used because it was the most suitable LLM from the available option for the specific evaluation is chosen. There are other LLMs like GPT-4 mini, GPT-4o etc., but due to free cost the Gemini was selected.

The evaluations were executed across the dataset. Allowing for the comprehensive testing without the need for manual annotation. This process enables to iterate quickly and refine the LLMs Prompts.

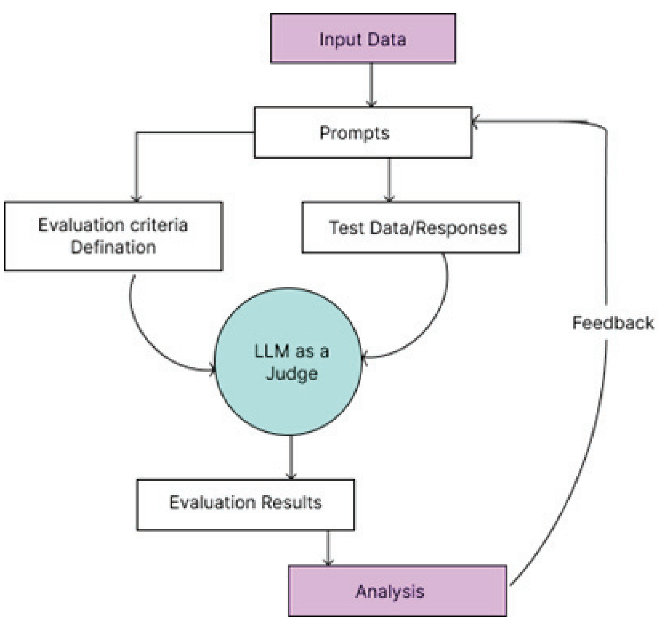

Fig. 2. LLM as Judge for Evaluation

TABLE I. EVALUATION CRITERIA

Evaluation Criteria	Description	Points
Keyword Presence (2 points)	Identify the presence of core concepts that should be included in a correct answer.	2 points for each keyword correctly mentioned.
Concept Accuracy (4 points)	Assess the accuracy and depth of the explanation provided by the student.	2 points: Detailed and accurate explanation. 1 point: Basic understanding with minor errors or lack of detail. 0 points: Incorrect or missing explanation.
Logical Structure (4 points)	Evaluate the organization and coherence of the answer.	3 points: Clear and logical organization with smooth flow of ideas. 2 points: Some organization but may lack a clear argumentative thread. 1 point: Limited or confusing organization.

IV. CHALLENGES AND LIMITATIONS

The automated grading system witness several challenges during its implementation. The major drawback observed while extracting text from the students answer sheets is the model deviates from the standard operating procedure of four iterations leading to inconsistencies in evaluations. Also, the complex handwriting styles of various student answer sheets makes it complicated for the model to evaluate in all cases. Re-prompting the model will produce accurate results by enabling self-correction. Recursive prompting works well but is not a universal solution.

The modern OCR technologies like Gemini 1.5 Pro struggle with low contrast lines and shades thereby affecting the visibility and accuracy of extracted text from answer sheets. In addition to this, the model fails to evaluate detailed answers by receiving only satisfactory responses which affects the score for the student answer sheet. The quality of extracted text is affected by noise and errors like misspelled words while implementing OCR. A plan-and solve zero-shot classification approach will infer a correct evaluation criterion and assess the answers accurately compensating the inability to grade the student answer sheets.

The initial preprocessing task of converting raw PDF files into images and optimizing image resolutions for accurate evaluation impact the computational efficiency of the model. To overcome this, the preprocessing and image enhancement procedures should be effectively implemented to obtain accurate results on variety of datasets. There is a need for further advancement in NLP-driven assessment tools to assess the student answers with clarity. Therefore, fine-tuning the self-learning algorithms and thoughtful implementation of all procedures is necessary to enhance the efficiency and accuracy of the automated grading system.

V. RESULTS

The automated answer sheet evaluation system uses Gemini 1.5 Pro to extract and analyze text, aiming to fix the problems faced by older methods such as Pytesseract, which has an inaccurate interpretation of handwritten text. The evaluation looks at digitized answer sheets, comparing them with the right answers can be seen in Figure 3 and figure 4. Gemini 1.5 Pro achieves a precision score of 0.865, indicating 86.5% accuracy in correctly identifying answers marked as correct by the model. Similarly, the system reveals a recall of 0.865, finding exactly 86.5% of actual correct answers, underscoring its capability to minimize false negatives. It reflects with a balanced F1-score of 0.865, demonstrating how well it can spot correct answers with minimal errors.

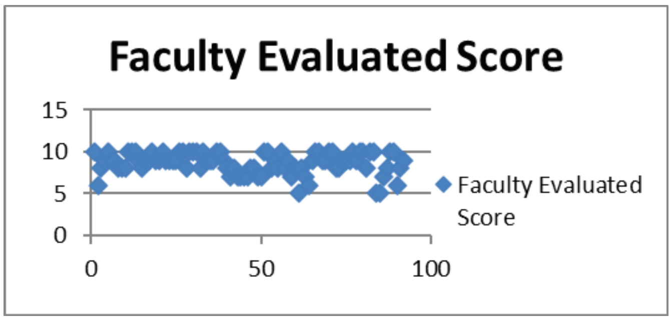

Fig. 3. Faculty Predicted Score

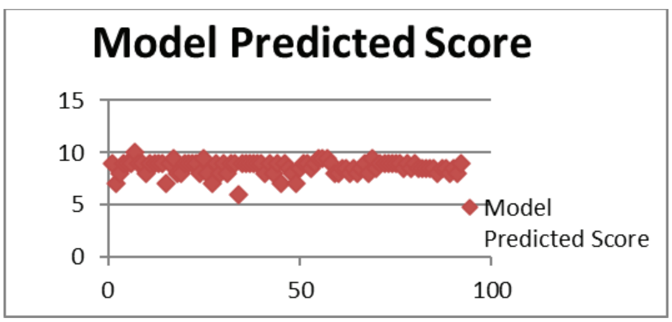

Fig. 4. Model Predicted Score

A. *Possible Reasons for Discrepancy:*

Faculty might notice good effort and potential in students, even if their answers are not perfectly accurate. This kind of understanding is tricky for NLP to capture. Implicit knowledge: Faculty see signs of student understanding that are not clearly stated in their answers.

Different criteria: Might be prioritized by the faculty, such as clarity, organization, or depth of reasoning, compared to the NLP model.

B. *Efficiency and Performance:*

The system is measured by seeing how it handles various image resolutions without slowing down or making mistakes. The interlacing of .png images does not adversely affect the processing time or accuracy. This approach seems to be more accurate and efficient than older methods. Gemini 1.5 Pro manages different writing styles well while also removing noise for better results. However, there are some instances when the model does not follow the four iterations principle, leading to inconsistent evaluations. These issues are resolved by re-prompting the model, which then leads to accurate results.

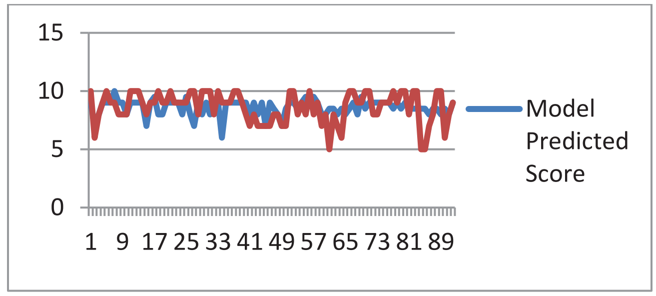

Fig. 5. Comparison of Model Predicted Scores and Faculty Evaluated Score

VI. ADVANCEMENTS AND FUTURE DIRECTIONS

This approach is best for extracting and evaluating text from various sources such as Images, PDFs, those can be Handwritten or digital transcripts. Utilizing AI and LLM models for the transcribing makes it simpler and faster than utilizing the conventional picture segmentation method. A few improvements are missing from this research. The extraction and analysis of tables and diagrams from the response sheets. It is necessary to make these developments in the extraction and evaluation of captions, diagrams, and tables.

User Interface and Integration: The current online examination method may be enhanced by the digital evaluation system and onscreen assessment technology, which can also lower the cost and simplify the process of determining the results. The mechanism for marking and scoring descriptive replies may be fully automated.

VII. CONCLUSIONS

This research work reduces the amount of human effort required for answering sheet evaluation and grading by utilizing LLM and AI technology. Despite the possible disadvantages of manually scanning an answer sheet, particularly when managing a high volume of sheets. However, there are ways to speed up the scanning process, such as high-speed professional scanners that can scan hundreds of pages in a minute or automated document feeders (ADF). The proposed approach offers a better way to automate the assessment of the answer booklet by attempting to extract and assess the subjective responses. This may be expanded even further to assess written material in many languages and provide students with constructive feedback for improved development.

ACKNOWLEDGEMENTS

Guides from the Society for data science are thanked for providing their assistance and direction during the research project. Grateful to Army Institute of Technology in Pune and Pragati Engineering College in Andhra Pradesh for providing the dataset utilized in this research.

REFERENCES

[1] K. Kazi *et al.*, "Trends of Artificial Intelligence for Online Exams in Education," *Artic. Int. J. Early Child. Spec. Educ.*, vol. 14, no. 01, pp. 2457–2463, 2022, doi: 10.9756/INT-JECSE/V14I1.290.

[2] V. Rowtula, S. R. Oota, and C. V. Jawahar, "Towards automated evaluation of handwritten assessments," *Proc. Int. Conf. Doc. Anal. Recognition, ICDAR*, pp. 426–433, 2019, doi: 10.1109/ICDAR.2019.00075.

[3] M. Kulkarni, G. Adhav, K. Wadile, R. Chavan, and V. Deshmukh, "Digital Handwritten Answer Sheet Evaluation System," pp. 0–16, 2024, [Online]. Available: https://doi.org/10.21203/rs.3.rs-3978232/v1

[4] H. Jain, M. Sherali Shaikh, R. Shankar, and V. Mishra, "Automatic Grading of Handwritten Answers," *Int. Res. J. Eng. Technol.*, no. May, pp. 409–415, 2022, [Online]. Available: www.irjet.net

[5] A. K. . Maya, J. Nazura, and B. L. Muralidhara, "Recent Trends in Answer Script Evaluation – A Literature Survey," *Proc. 3rd Int. Conf. Integr. Intell. Comput. Commun. Secur. (ICIIC 2021)*, vol. 4, no. Iciic, pp. 105–112, 2021, doi: 10.2991/ahis.k.210913.014.

[6] M. A. Hussein, H. Hassan, and M. Nassef, "Automated language essay scoring systems: A literature review," *PeerJ Comput. Sci.*, vol. 2019, no. 8, pp. 1–16, 2019, doi: 10.7717/peerj-cs.208.

[7] M. F. Bashir, H. Arshad, A. R. Javed, N. Kryvinska, and S. S. Band, "Subjective Answers Evaluation Using Machine Learning and Natural Language Processing," *IEEE Access*, vol. 9, pp. 158972–158983, 2021, doi: 10.1109/ACCESS.2021.3130902.

[8] A. D. Alrehily, M. A. Siddiqui, and S. M Buhari, "Intelligent Electronic Assessment for Subjective Exams," pp. 47–63, 2018, doi: 10.5121/csit.2018.80804.

[9] V. Bahel and A. Thomas, "Text similarity analysis for evaluation of descriptive answers," 2021, [Online]. Available: http://arxiv.org/abs/2105.02935

[10] L. Wang *et al.*, "Plan-and-Solve Prompting: Improving Zero-Shot Chain-of-Thought Reasoning by Large Language Models," *Proc. Annu. Meet. Assoc. Comput. Linguist.*, vol. 1, pp. 2609–2634, 2023, doi: 10.18653/v1/2023.acl-long.147.